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The Shapes of Clines and Wavefronts

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ABSTRACT

Cline theory has a central place in speciation studies. Cline locations delimit taxon boundaries, cline widths scale with barrier strength, and the shapes of clines (smooth or stepped) suggest whether species barriers are mono- or polygenic. How cline shapes vary along chromosomes therefore forms part of the genome's species barrier landscape. Further, asymmetric moving clines (wave fronts) can mark adaptive introgression puncturing species barriers, potentially leading to their collapse or decay. Here we review the development of cline and wavefront models and relate this to the use of dispersal kernels in epidemiology and ecology. We contrast classical results to those for a thick-tailed kernel, showing how cline shape affects the speed of spatial processes, including the widening of neutral clines and the spatial coalescent. We critique current cline models used for inference (both spatial and genomic clines) and address Barton's question: Why (after decades of cline fitting) is there so little evidence of stepped clines? We suggest evidence is weak because stepped cline models are over-parameterised, while current genomic cline models are under-parameterised. We explore goldilocks cline parameterisations and discuss non-parametric approaches that may help resolve these issues. This broadens to a discussion of the future of, and alternatives to, cline fitting.

1 | Introduction

> Those forms which possess in some considerable degree the character of species, but which are so closely similar to some other forms, or are so closely linked to them by intermediate gradations, that naturalists do not like to rank them as distinct species are in several respects the most important to us.
>
> (Darwin 1859)

> Although the early writers thought a good deal about the effects of geography and dispersal [...], intense geographic differentiation and speciation was thought to require some kind of island or complete isolation situation. Only in the last few years have population geneticists become seriously concerned with the effects of gene flow in continuous populations.
>
> (Endler 1977, 3)

> How many genomic regions differentiate during speciation? How small are regions where divergence significantly exceeds the genomic average ([...])? How are regions of exceptional divergence dispersed around the genome? We suggest that recent discussions of these issues in the context of ecological speciation would benefit from closer attention to well-established cline theory.
>
> (Abbott et al. 2013)

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Glossary (Concepts in approximate order of first occurrence)

PDF: Probability density function.
CDF: Cumulative distribution function. In 1D: convolution of a unit step function with a PDF.
Dispersal kernel: The contribution of a specific core (source point, parent, parental copy) to the re-organisation of certain units (e.g., offspring) in a larger entity (population). See (Nathan et al. 2012) for history and proposed usage in ecology and evolution. Herein 'dispersal kernel', unless otherwise stated, will mean dispersal location kernel.
Dispersal location kernel: The PDF for the end location of a parent-offspring dispersal vector. In an n-dimensional field area, this is an n-dimensional PDF (See Box 1, first pane). The Normal dispersal location kernel has a zero-centred bell shape in any dimension.
Dispersal distance kernel: The PDF for distance covered by a parent-offspring dispersal vector (see Box 1, second pane). No information is lost when summarising a radially symmetrical dispersal *location* kernel in any dimension as a dispersal *distance* kernel of one dimension. The Normal dispersal distance kernel is a half bell shape in 1D, but this shape changes as the mode shifts away from zero in higher dimensions.
Effective dispersal kernel: the dispersal location kernel as observed after the effects of selection.
Dispersal kernel scale: In 1D, the x-axis distance measure that gives unit area under the curve of a density function, as required for a PDF. Here we assume the distance scale is equal along all axes in higher dimensions.
Shape of a distribution: All moments of a distribution other than the first two, location and scale. This leaves skewness (asymmetry), kurtosis (thick-tailed-ness), and further moments with increasingly subtle descriptions. Kurtosis exceeding that of the Normal distribution is leptokurtosis.
Thick-tailed (also fat-tailed): Of distributions or clines, tails thick compared to central change (Figure 1d). See also shape of a distribution and shape of a cline.
Cline width: The inverse of the maximum gradient of a smooth change in trait.
Cline centre: The turning point of maximum gradient of a smooth change in trait.
Shape of a cline: By analogy with the shape of a distribution, all aspects of a cline other than centre and width (analogues of location and scale), in particular asymmetry and stepped-ness (analogues of skew and kurtosis). Leptokurtotic kernel distributions give rise to stepped clines, Figure 1c,d.
Geographic cline: trait change along geographic distance x-axis.
Genomic cline: trait change along hybrid index x-axis.
Sigmoid clines: Of geographic clines: S-shaped. A term used for both neutral clines (Neutral, Table 1, Figure 1b) and clines where selection acts without recombination (Logistic, Table 1). When the hybrid index is substituted for distance on the logistic cline x-axis, the resulting genomic cline has a logit-logistic shape (equation 4b in Fitzpatrick 2013), Box 3).
Stepped clines: In comparison to a sigmoid cline, or its genomic equivalent, the central portion is steepened (Figure 1d). A prediction of multilocus clines interacting (Szymura and Barton 1986). When allele frequencies at selected loci change in concert, each experiences its own direct selection, but also indirect selection through associations across loci, steepening central clinal change. Stepped-ness can be measured by fitting a tripartite cline model (Figure 2), or introducing a shape parameter for stepped-ness.
Unit cline: A zero-centred cline of unit width (unit central gradient). Distinct unit clines differ only in shape. Any unit cline $U(x)$ can be {recentered, rescaled} to any {centre, width} $\{c, w\}$ as $U\left(\frac{x-c}{w}\right)$. See unit time clines in Figure 1, black and grey zone clines in Box 1.
MAD: Maximum absolute difference: a comparison of two functions across a set of points. Here, unless otherwise stated, unit clines are compared for points $-4 \leq x \leq +4; \Delta x = 0.01$, that is, over 8 cline widths.
Stable distributions: A distribution is stable if a linear combination of two independent random variables with this distribution has a distribution of the same shape, that is, differing at most in location and scale parameters.
Convolution of a distribution: A linear combination of n independent random variables with this distribution has a distribution described by its n-fold convolution. The two-fold convolution of 1D continuous dispersal kernel $k(\cdot)$ is $k_2(x) = \int k(z).k(x-z)\,dz$ (note that for all values of z the sum of the arguments of k is x). Only stable distributions do not change shape under convolution. Only a small proportion of distributions have analytic solution under convolution, however *convolution is simple to approximate to arbitrary accuracy by simulating (large) arrays of random variates and adding them*.
Indirect selection: Change in allele frequency at one locus due to selection acting at another locus (or loci) in statistical association (linkage disequilibrium). As admixture generates linkage disequilibrium genome-wide, *indirect selection will affect unlinked loci*. Calling indirect selection 'linked selection' is therefore unnecessarily confusing (Stankowski et al. 2019).
Exogenous/endogenous (extrinsic/intrinsic) selection: The cause of exogenous selection is tied to the environment (e.g., taxa or genes are adapted to different regions or niches). Endogenous selection is caused by genome interactions independent of the environment. See (Kruuk et al. 1999) on expectations for clines maintained by endogenous versus exogenous selection.
Tension zone: A hybrid zone maintained by endogeneous selection, and thus free to move across the environment. Tension zones move down density gradients to become trapped in density troughs or at physical barriers to gene flow (Barton 1979a).

The first mathematical treatment of migration and selection in continuous populations was by R.A. Fisher (Fisher 1937), who studied the wave of advance of favourable genes. The next year Huxley coined the term 'cline' (Huxley 1938), and a decade later, by allowing selection coefficients to depend on location, Haldane (Haldane 1948) developed an equilibrium model of gene flow and selection in a cline, closely related to Fisher's model (Nagylaki 1975), and used it to estimate the intensity of

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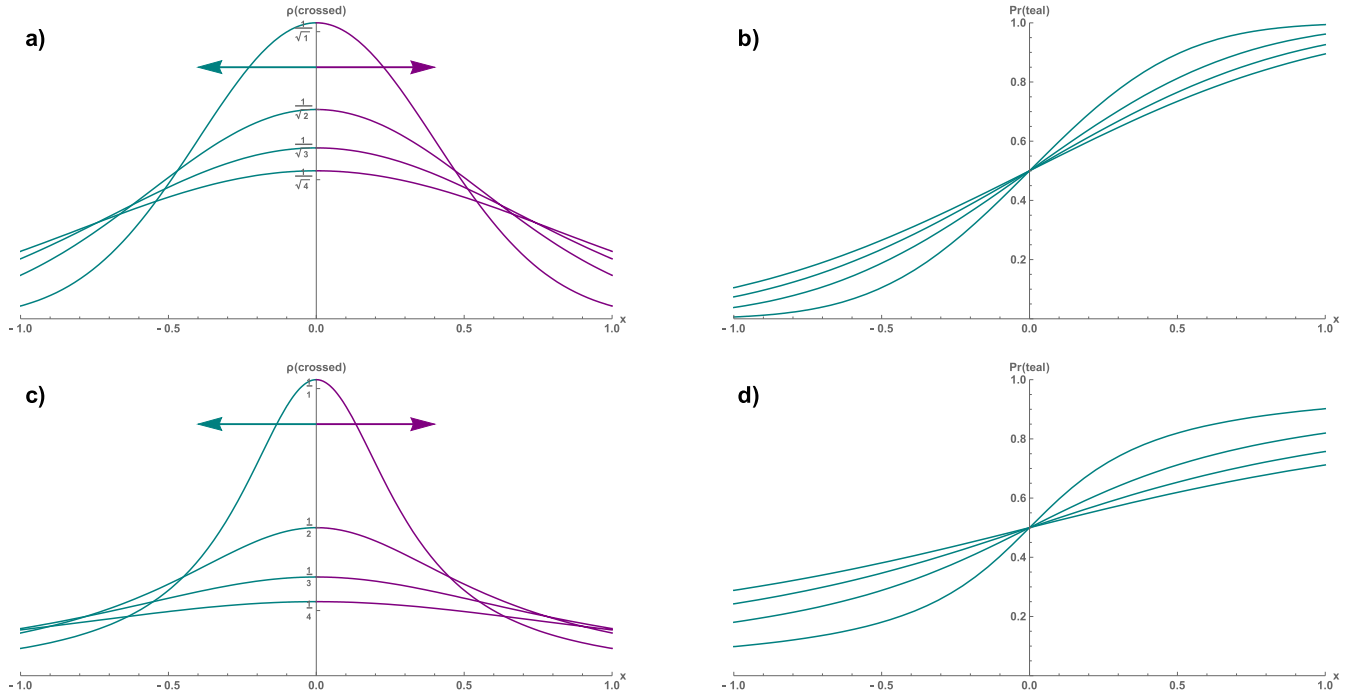


FIGURE 1 | The neutral contact case for stable dispersal kernels over four discrete generations: Two populations (purple arriving from x −ve, teal from x +ve) meet centrally and (left panes) their alleles spread across the contact following a dispersal location kernel. Top panes: When the kernel is (a) the Normal distribution PDF, the expected neutral cline is (b) the 'sigmoid' Normal Distribution CDF. The Normal PDF (a) can be convoluted over generations to give the characteristic widening bell-shaped curves of the diffusion of Brownian particles. The Brownian cline width (b) increases with the square root of time, as with the displacement of a Brownian particle (Einstein 1905). Bottom panes: When the kernel is (c) the thick-tailed Cauchy distribution PDF, the expected neutral cline is (d) the 'stepped' Cauchy Distribution CDF. The Cauchy PDF (c) can also be convoluted over generations to give widening leptokurtotic curves. The neutral Cauchy kernel cline width (d) increases linearly with time, faster than Normal. All panes: Initial PDFs are scaled (see Table 1) such that CDF clines have unit width at unit time. (Right panes) Cline widths are the inverse of the central gradients of the CDFs, and so by definition equal to the denominators of the central values of the PDFs (left panes). Only rising (teal) clines are shown; the complementary (purple) falling clines are redundant, and omitted for clarity.

natural selection in deer mice. Thus wavefronts, clines, and inference from them, lie at the roots of the modern evolutionary synthesis, and here we will use cline theory as a catch-all for this body of work. While (Nagylaki 1975) was setting out to "relate clearly the theory of clines to the diffusion methods [...] which have been very productive in population genetics", tropical ecologists were formulating a very different treatment of migration and selection: the Janzen-Connell model (See Terborgh 2020) arose from the empirical observation that seedfall is most concentrated around fruiting trees, whereas sapling recruitment fails close to parent trees and succeeds at a distance. These and analogous empirical observations inspired the use of dispersal kernels to summarise dispersal probabilities in ecology (Nathan et al. 2012) epidemiology (Pybus et al. 2012) and invasion biology (Kot et al. 1996; Lindström et al. 2011).

It would seem then that dispersal kernels should also appear at the heart of cline theory, but this was not to be. This is because the diffusion method made famous by Einstein's description of Brownian motion (Einstein 1905), gives rise to an *implicit* dispersal kernel: the Normal distribution. The first part of Einstein's argument was to determine how far a Brownian particle travels in a given time interval. Assuming many small random steps, he found the density of particles at a given time satisfies a diffusion equation, the solution of which is the Normal distribution, a stable distribution which widens over time, changing scale without changing shape (Figure 1a). In this way Einstein demonstrated that the displacement of a Brownian particle increases with the square root of time (Einstein 1905). The mission of Nagylaki and others to "relate clearly the theory of clines to the diffusion methods" (Nagylaki 1975) tied early spatial genetics to the implicit Normal dispersal kernel of the Brownian particle at the same time as ecologists were confirming early suggestions (e.g., Bateman 1950) that biological dispersal often differed from that of a Brownian particle, with thicker tails than Normal (leptokurtotic). Despite discussion of a diversity of circumstances (e.g., Wright 1969; Diekmann 1978; Nichols and Hewitt 1994), the parent-offspring displacement distribution, keystone of spatial population genetics, has almost always been described only in terms of a variance (a proxy for scale, explored later) and so has been implicitly assumed to only ever have one shape: the Brownian Normal (Bateman 1950; Nathan et al. 2012). But parent-offspring dispersal need not conform to the notion of many small and exchangeable random steps (Petrovskii and Morozov 2009); normal is just one dispersal kernel—other kernel shapes are possible, and kernel shape makes a difference (Richardson et al. 2014).

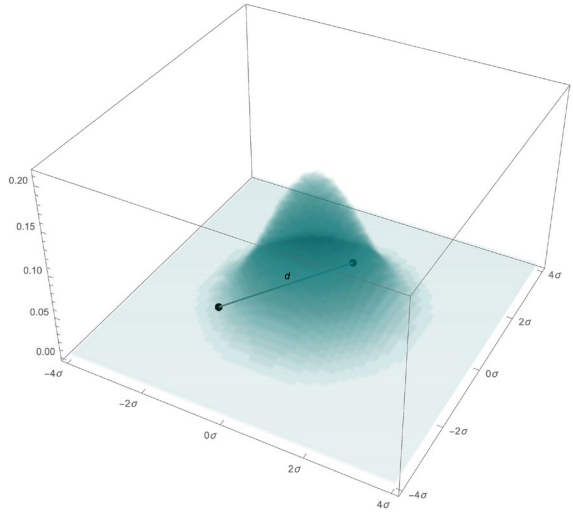
2 | Neutral Clines

The effect of dispersal kernels on cline shape can be illustrated in the neutral case by contrasting two stable distributions for parent-offspring displacement: Normal versus Cauchy (with thicker

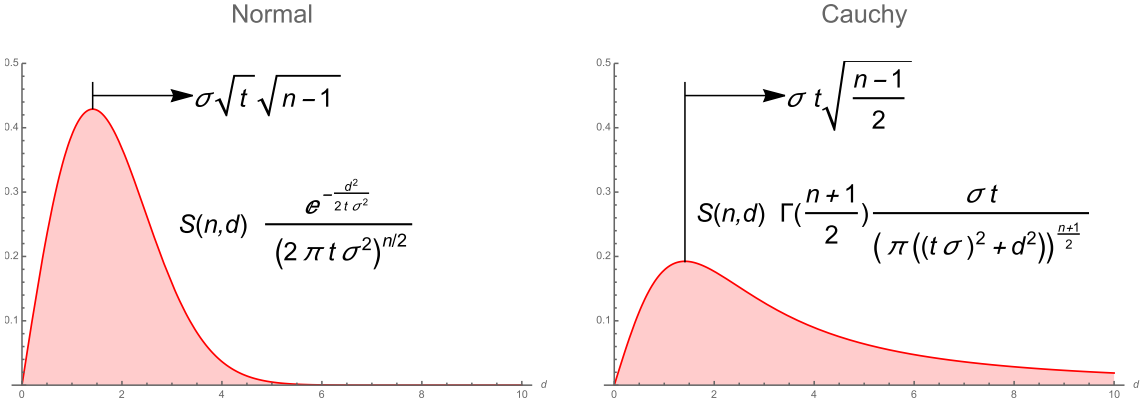
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BOX 1 | Scale, speed and neighbourhood size.

Scale: What are the natural units of cline width? Most biologists would suspect a 1 km wide hybrid zone (HZ) is narrow for birds, but wide for snails, that is, a cline is not narrow or wide based only on its width in SI units. Instead we use units specific to the study organism: the per generation scale of dispersal σ (a length, smaller for snails, larger for birds). Setting the origin of a frame of reference at a parent, a zero-centred radially symmetrical multivariate PDF can be used to define a dispersal *location* kernel K with scale σ for offspring: Here (grey green shaded PDF), the field area has $n = 2$ dimensions (e.g., {x,y}) measured in σ, and two offspring of the same parent are joined by a line length d. The 1 km HZ is narrow for birds if their σ is $\gg 1$ km, and conversely wide for snails if their σ is $\ll 1$ km. If we measured bird and snail HZs in 'natural' σ units, and they had similar histories, governed by similar processes, then they would have similar width. This is how cline theory 'scales' over diverse study systems.



Speed: The distance d between two offspring of the same parent follows the dispersal *distance* kernel found by sampling two location vectors from K and adding them to derive K_2, the two-fold convolution of K. For the stable distributions in Figure 1, the t -fold convolution can be expressed for any dimension n of field area using $S(n, d)$, the surface area of the n-sphere radius d.



The pink shaded dispersal *distance* kernel PDFs are $S(n, d)K_t(n, d)$ for the respective kernels, and are plotted for $n = t = 2$. The position of the modal distance (black verticals) shows us how the spatial scale of a chain of inheritance increases over time: linearly for the Cauchy kernel, but with the root of time for the Normal kernel, generalising the (Figure 1) observation that different kernel shapes give different speeds of process, from the widening of clines in 1D, to spatial inheritance in any dimension of field area. This has consequences for the rate of coalescence.

Neighbourhood size: As the probability of pairwise coalescence in the previous generation, in an idealised population without spatial context, is $\frac{1}{2N}$, so in an idealised spatial population, for the two offspring above, this coalescence probability is $\frac{1}{\rho}K_2(n, d)$. N is population size, ρ population density (per spatial extent) and $K_2(n, d)$ expresses how pairwise parent sharing probability falls off with distance d in n dimensions (Baird and Santos 2010). We can remove the dependance on distance by letting d tend to zero (offspring found in contact). Then, for a 2D field area and Normal dispersal, this coalescence probability is $\frac{1}{4\pi\rho\sigma^2}$, the inverse of Wright's neighbourhood size. Neighbourhood size changes smoothly with the shape parameter of the 2D Students-t dispersal kernel as $(1 + \nu)/\nu\, 4\pi\rho\sigma^2$: the rate of coalescence is halved to $\frac{1}{8\pi\rho\sigma^2}$ by the (Figure 1) shape change between Normal and Cauchy cases. Neighbourhood size is also affected by the dimension n of the dispersal kernels as $2^n\pi^{n/2}\rho\,\sigma^n$ (Normal) and $2^n\pi^{(n+1)/2}\rho\,\sigma^n\,\Gamma\left(\frac{1+n}{2}\right)$ (Cauchy). Note 1: Here, the spatial scale σ is always raised to the dimension of spatial extent, whereas the density is always measured *per* spatial extent. Spatial extent cancels when they are multiplied, leaving neighbourhood size a non-spatial quantity commensurate with population size N. Note 2: The Normal PDF is special in that its variance is equal to its scale squared: σ^2. This is not a general property of PDFs: where other dispersal kernels are considered, the variance is not a useful proxy for scale.

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tails), and the clines they generate (Figure 1). As with the displacement of a Brownian particle, the width of a Brownian cline increases with the square root of time (Figure 1a,b). In contrast, the width of clines for the thick-tailed Cauchy kernel increases faster—linearly with time (Figure 1c,d). If we were going to infer time since neutral contact between populations from the widths of clines in traits, we could reduce our error by co-estimating the shape of the clines because the shape could indicate whether we are dealing with a fast or slow spreading kernel (e.g., Figure 1c,a respectively). If we were going to infer the existence of a polygenic species barrier from stepped clines, we should be aware that a thick-tailed dispersal kernel can give stepped (thick tailed) clines even in the neutral case (Figure 1d). But we are getting ahead of ourselves: Figure 1 shows neutral clines which are CDFs of stable location dispersal kernels, and the only selection mentioned so far is that of Fisher's wave of advance of an advantageous gene. While that work gave rise to its own entire field of endeavour (Invasion Biology, Skellam 1951; Kot et al. 1996; Lindström et al. 2011; Phillips 2025), it is selection against admixture which has been key to understanding the body of cline theory most entangled with speciation: hybrid zone clines.

3 | Selection Against Admixture

Can selection against genome admixture arrest the ever-widening neutral clines of Figure 1b,d to produce a stable 'gradation' between populations of the sort Darwin envisaged in the opening quote? For the Normal dispersal kernel, diploid genomes, no recombination, and selection against F1s, Bazykin showed the answer is *yes* (Bazykin 1969). Bazykin's selection-stabilised clines can also be described in terms of free recombination with selection against heterozygotes at one single locus. This might seem a simpler description, but it must be caveated that no indirect selection is acting. Because admixture generates genome-wide associations across loci (S. J. Baird 2015), potential sources of indirect selection could lie anywhere in the genome, and so the Bazykin cline, when framed in terms of heterozygotes, is an expectation for the causal locus of a *monogenic* species barrier. In both these framing of the Bazykin result, only one non-recombining genome region is under selection, and only three genome types are distinguished: two pure and one eqi-admixed (a heterozygous diploid locus or an entire F1 individual's genome). Selection against the admixed type arrests cline spread and distorts the equilibrium cline shape from that of the neutral Normal CDF to that of the CDF of the logistic distribution (See Table 1). Despite this change in shape both Normal and logistic clines are called sigmoid. The Bazykin result applies at two recombination extremes, and it is useful to bear both in mind, for while a species barrier maintained by selection against heterozygotes at a single locus may sound infeasible, one maintained by selection against F1 individuals, genome wide, may not. At the F1 extreme the potential for indirect selection is maximal because *cis* associations between all genes in diverged genomes are maintained, while every site of the genome is heterozygous by source. Then, even in the presence of recombination, strong selection against F1-like individuals will reduce the effective recombination rate, slowing the decay of pure *cis*-genome associations, and increasing the potential for indirect selection at multiple loci to further distort cline shape. The paths of these initial changes in multilocus cline shape, dependent on both hybrid indices and degree of heterozygosity by source, are spatially explicit cousins of the paths taken through Fisher's geometric fitness space during admixture (Simon et al. 2018): whereas in a panmictic population admixture genotypes meet, mate and compete only as a function of their relative frequencies, in a spatially explicit context these evolutionary events also depend on relative location. For example, a stepped cline reduction in effective recombination rate will preferentially affect genotypes that are common centrally. Returning to the no-recombination case, Bazykin not only demonstrated a spatial dispersal-selection equilibrium existed, but also calculated, to a weak selection approximation, the expected width of his clines, which is $\sigma\sqrt{8/s}$, s being the selection acting against heterozygotes and σ, the scale of parent-offspring dispersal (see Box 1). With hindsight, and on considering the effects of kernel shape, we see the expected width of a cline maintained by Bazykin selection will depend not only on the selection acting, but also on the dispersal kernel shape.

While (Nagylaki 1975) showed that Haldane's (1948) exogenous cline could be recast as a special case of a Fisher wavefront, it was Barton (1979a), section 3: (i) who pointed out that there is a family of solutions between Bazykin's endogenous symmetric cline and Fisher's (endogeneous) asymmetric travelling wave, depending on whether two pure genome types are equally fitter than their admixed type. Equality gives the symmetric Bazykin cline; inequality, travelling clines moving toward the less fit pure type. Barton (1979b) also showed that selection on multiple loci (a polygenic barrier) distorts cline shape through indirect selection, steepening the cline centre relative to its tails to produce stepped clines. In fact, in those two seminal 1979 papers. Barton laid out the properties of spatial genome admixture as we understand it today, and went on to show, with Hewitt (Barton and Hewitt 1985), that the majority of hybrid zones had cline widths too narrow to be consistent with maintenance at the scale of environmental change (as explored by Haldane), and so instead were likely to be tension zones maintained by endogenous barriers. Here we see how deeply species and cline concepts can become entangled, as it is tempting to define species as only those taxa kept distinct by intrinsic barriers (all other barriers being context dependent, and therefore potentially ephemeral). Arguments for a more operational taxonomy (e.g., Mallet 1995) have largely fallen on deaf ears, although it does appear acceptable to describe multilocus endogenous barriers as species barriers irrespective of how taxonomists rank the organisms on either side (see Krieber and Rose 1986, e.g., *Mus* 'subspecies', Albrechtová et al. 2012).

With (Barton 1979a) came the possibility of inferring the existence of a multilocus species barrier (polygenic selection) by deciding whether or not an observed cline had a steepened central portion. (Barton and Bengtsson 1986) developed a continuous explicit model of stepped cline shape, and (Kruuk et al. 1999) went on to show this stepped shape was similar whether multilocus selection was endogeneous or exogeneous. The Kruuk result is not for one cline shape, but rather a continuum of multilocus stepped shapes with the Bazykin shape at the single locus limit. In the meantime, (Gavrilets 1997) had shown how viability selection, even acting on a single locus, can also result in stepped cline shapes. As with Kruuk, the Gavrilets result is for a continuum of stepped shapes, Bazykin-shaped at one extreme of the selection model, but with the CDF of the Student's-t distribution (with 2 degrees of freedom, or shape parameter 2) at the other extreme (Gavrilets 1997, equation 14a). We will use $T(2)$ to refer to the unit cline with the shape of this CDF (see Table 1).

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TABLE 1 | Symmetric *UnitClines* and their inverses. 'Scale': For distribution φ, $CDF[\varphi(Scale), x] = UnitCline_{\varphi}[x]$. Blue: Alternate forms. *MAD* Maximum absolute difference (between a *UnitCline* and the closest *T*-cline). Key to functions: $\{Erfc, Erfc^{-1}\}$ the complementary error function and its inverse; *Tan* tangent; *Log* natural logarithm; *Tanh* Hyperbolic tangent; *Sgn*[*x*] the sign function; $\{I_z, I^{-1}\}$ the incomplete beta function and its inverse $\left(I^{-1}\left[I_z[a,b],a,b\right] = z, z^{-1}(z(x)) = x\right)$; β the beta function.

Context	PDF Scale	*UnitCline* CDF; $\{c,w\} = \{0,1\}$ $p = U(x)$	Inverse *UnitCline* $x = U^{-1}(p)$	Cline names $T(\cdot)$
Many small random steps between parent and offspring	Normal $\sqrt{2\pi}$	$Erfc[-\pi x]/2$	$-Erfc^{-1}[2p]/\sqrt{\pi}$	Brownian $T(\infty)$ $\approx T(116)$ *MAD* 0.001
Thick-tailed dispersal kernel	Cauchy π	$\frac{1}{2} + Tan^{-1}[\pi x]/\pi$	$Tan[\pi(p-1/2)]/\pi$	Cauchy $T(1)$
Selection against admixture of a single non-recombining genome region	Logistic 4	$\frac{1}{1+e^{4x}}$ $1/2 + Tanh[2x]/2$	$Log\left(\frac{p}{1-p}\right)/4$ $Tanh^{-1}[1-2p]/2$	Bazykin $\approx T(6.67)$ *MAD* 0.001
Viability selection against admixture of a single non-recombining genome region	Student-*t* (shape 2) $2\sqrt{2}$	$\frac{1}{2} + \frac{x}{\sqrt{1+4x^2}}$	$Sgn[p-1/2]\frac{\sqrt{4p-1-4p^2}}{4\sqrt{p^2-p}}$	Gavrilets (stepped extreme) $T(2)$
Model of symmetric clines	Student-*t* (shape v) $\beta\left[\frac{v}{2},\frac{1}{2}\right]\sqrt{v}$	$\frac{1}{2}I_{z(x)}\left[\frac{v}{2},\frac{1}{2}\right]$; $x<0$; $1-\frac{1}{2}I_{z(x)}\left[\frac{v}{2},\frac{1}{2}\right]$; $x\geq 0$; $z(x) = \frac{1}{1+x^2\beta\left[\frac{v}{2},\frac{1}{2}\right]}$	$z^{-1}\left(I^{-1}\left[2p,\frac{v}{2},\frac{1}{2}\right]\right)$; $p<\frac{1}{2}$; $z^{-1}\left(I^{-1}\left[2(1-p),\frac{v}{2},\frac{1}{2}\right]\right)$; $p\geq\frac{1}{2}$; $z^{-1}(\omega) = \frac{\sqrt{1-\omega}}{\beta\left[\frac{v}{2},\frac{1}{2}\right]\sqrt{\omega}}$	*T*-cline $T(v)$

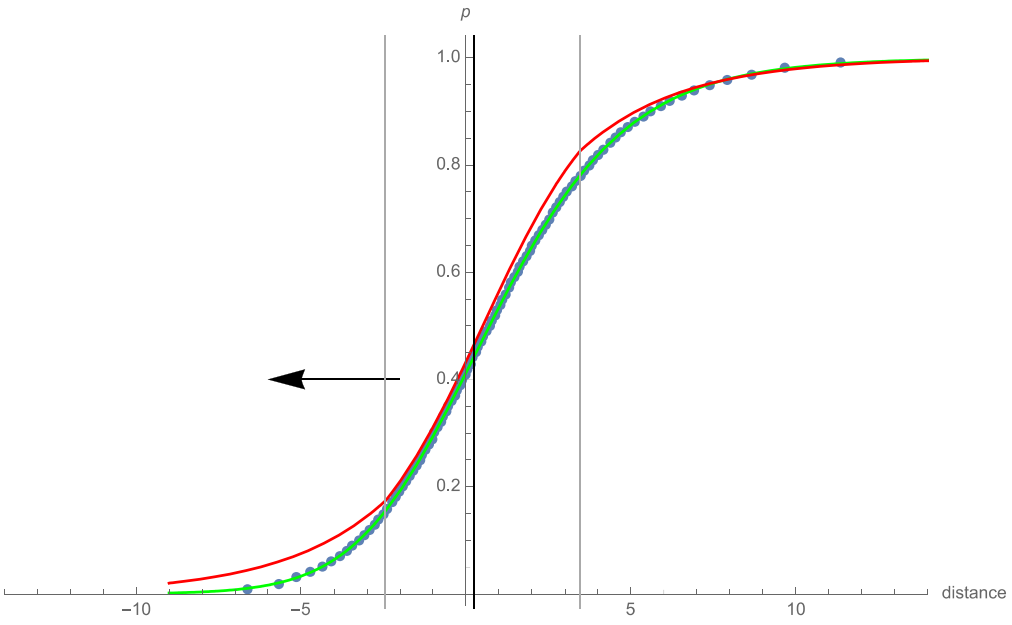


FIGURE 2 | Fitting models to the shape of Fisher's wave of advance. Blue points: Fisher tabulated values for the wave of advance in table IV (Fisher 1937). This is an asymmetric cline of {centre, width} $\{c,w\} = \{0.256, 8.110\}$ as estimated from Fisher's table. The centre is marked with a black vertical line. Green line: Degrees of freedom 8.365 and non-centrality 1.367, parameters of the Student-*t* distribution PDF, can be used respectively to shape stepped-ness and asymmetry of the cline $T\left(\frac{x-c}{w}, 8.365, 1.367\right)$ that differs nowhere by more than 0.001 from the Fisher values (MAD 0.001). Red line: A tripartite cline fit to the Fisher data. Vertical grey lines mark joins between the central logistic part and exponential tails. The right hand join is shown for the best fit, the best fit left hand join falls further to the left than shown (out of frame), but least squares fitting with $\{c,w\}$ and all four shape parameters (two for each tail) free to vary only achieves MAD 0.018, due to a poor match at the best fit 'corner' marked by the right hand grey vertical.

All of these selection cline results assume the Normal dispersal kernel implicit in the diffusion method. Together with the previous section we now see that stepped cline shapes can be expected with or without selection (given variation in dispersal kernels), and with or without multiple loci (for Normal dispersal kernels, given a diversity of selection regimes). What then can we hope to infer if we observe a stepped cline shape? Further: Neither the Kruuk nor the Gavrilets stepped result is even expressed as

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a cline function that could be fitted to data—this is why they have no entries in Table 1. The Kruuk result (Kruuk et al. 1999, equation 14) is in the form of an ordinary differential equation parameterised by a coupling coefficient $\phi = (L-1)\frac{s}{r}$ of uniform coupling coefficients $\frac{s}{r}$ summed over the interactions between L loci. Instances of the equation can be numerically solved using the NDSolve tool in Mathematica (Wolfram Research 2019). The Gavrilets result (Gavrilets 1997, equation 14b) takes the form of the inverse function of a cline $g^{-1}(C,p) = x$ which, unlike the functions in Table 1, itself has no obvious inverse. Instead Gavrilets clines can be numerically approximated by tabulating $\{g^{-1}(C,p), p\}$ over values of p and interpolating. This means that, in contrast to sigmoid clines, there is no straightforward way to plot known results for stepped clines, prompting the question: when we talk of stepped clines, what is our model?

4 | Cline Models and Inference

While Haldane himself said he lacked sufficient data to support estimates of cline parameters for deer mice, he dedicated a large section of his discussion on how data hungry such estimates are (Haldane 1948). An exemplary field sampling effort and allozyme allele counting allowed estimates of the parameters of a stepped cline between *Bombina* subspecies (Szymura and Barton 1986, 1991), a model-based analysis that set the paradigm for hybrid zone inference software for decades (Analyse (Baird and Barton 1995), ClineFit (Porter et al. 1997), Cfit (Gay et al. 2008), HZAR (Derryberry et al. 2014)). That original stepped cline model is a tri-partite composite (see Figure 2) of a 'sigmoid' (logistic) central portion joined to exponential tails (Szymura and Barton 1991). There are four shape parameters corresponding to a barrier strength and a tail decay rate in either direction. This allows for both cline asymmetry and central steepening using parameters that have direct interpretation for evolutionary process. Further, likelihood comparison with simpler nested models (where parts of the parameter vector are fixed) allows powerful likelihood ratio tests for asymmetry and stepped-ness (e.g., Macholán et al. 2007). The inference framework built around the tripartite model allowed stepped clines to be identified in several further field systems, *Podisma* (Barton and Gale 1993), *Pontia* (Porter et al. 1997), *Mus* (Macholán et al. 2007), but perhaps not as many as expected under null (multilocus, polygenic) models of speciation (Barton and Charlesworth 1984): the long periods of reduced geneflow thought characteristic of the majority of speciation events imply incompatibilities accumulating at many loci. With increased geneflow these are expected to interact via indirect selection to steepen clines centrally. Thirty years after demonstrating the *Bombina* zone was stepped, Barton commented on the paucity of further examples "This may be because dense spatial sampling is needed to identify a step, but more likely is because the genetic map is typically long enough that selection does not often maintain a strong barrier." (Barton 2020). The first potential explanation is the data hunger noted by Haldane, the second refers to the balance between selection against admixture, and recombination, which admixes genomes (hence the coupling coefficients of the Kruuk result). Recombination breaks down the linkage disequilibrium generated by admixture and/or epistasis, weakening indirect selection and opposing epistatic selection. A given amount of selection against admixture might then be overwhelmed by a long genetic map (high recombination). While it is clear how this applies when selection and recombination are each described by a single parameter (Barton 1983; S. J. E. Baird 1995; Kruuk et al. 1999), it is less clear when both selection and recombination densities vary along the genome (Martin et al. 2019; Stankowski et al. 2019). Here instead we explore a third potential explanation for the paucity of stepped cline observations. Sampling data is not the only thing needed to identify a step: one also needs the model of what a stepped cline is, and how that differs from a non-stepped 'sigmoid' cline. The more free parameters this stepped model has, and the greater the distance between its expectations and the process generating the observations, the less power available to infer a step.

The flexibility of the tripartite stepped model is in natural tradeoff against both high parameter numbers and distance from 'reality'. It has four free parameters for asymmetric stepped shape whereas the analogous distribution-shaping moments are only two: skewness and kurtosis; it also has two discontinuities where its parts join, corners that are not a feature of the underlying expectations (Fitzpatrick 2013) (Contrast the 2-parameter T-cline fit to a 4-parameter tripartite fit in Figure 2).

5 | Stepped Clines With Fewer Parameters Make Powerful Models

We have shown a relationship between dispersal kernels (PDFs) and clines (CDFs). Where dispersal kernel PDFs have location and scale, clines have centre and width. Now we note this relationship extends to shape parameters: where PDFs have skewness and kurtosis, clines have (a)symmetry and stepped-ness. Our intention is to replace the four shape parameters of the tripartite cline model (which allow for varying asymmetry and stepped-ness) with just two parameters. Dealing first with stepped-ness: the symmetric clines of Figure 1b,d are sigmoid and stepped (thick-tailed) respectively, but their PDFs (1a,c) lie on a stepped-ness *continuum* parameterised by the shape parameter ν of the Student-t distribution, which describes both the Cauchy distribution at $\nu = 1$, and the Normal distribution for high ν (effectively indistinguishable from Normal for $\nu > 30$). Student-t parameter ν acts as a 'slider' for the variation in stepped-ness shown in Figure 1, but also allows for even more extreme stepped-ness (thick tails) than Cauchy for $0 < \nu < 1$. Further, at $\nu = 2$ this smoothly sliding stepped-ness scale includes Gavrilets' result for a single locus cline at its most stepped extreme (Table 1). We will call this {Cauchy, Gavrilets, Normal} inclusive family of clines T-clines, and denote these three instances $T(1)$, $T(2)$ and $T(\infty)$ respectively. Dealing second with (a)symmetry: the non-central Student's-t distribution can be re-expressed (For this and other mathematical details, see the Supporting Information) with an asymmetry parameter α, giving a plane of cline shapes $T(\upsilon, \alpha)$ from Brownian neutral sigmoid to stepped, and from left biased ($\alpha < 0$) through symmetric ($\alpha = 0$) to right biased ($\alpha > 0$).

This T-cline model space reduces the four tripartite cline shape parameters to two, has no discontinuities and includes both the neutral case and the stepped extreme of the Gavrilets continuum. $T(\upsilon, \alpha)$ as a model of the Fisher wave is shown in Figure 2: the maximum absolute difference (MAD)

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between $T(8.365, 1.367)$, for any value in (Fisher 1937) table IV is <0.001. This is in contrast to the best fit tripartite model at MAD 0.018. For the Fisher wave the T-cline model attains this 18-fold increase in accuracy with two less free parameters than the tripartite model, suggesting much more power is available for inference. All these features of the T-cline as a shape model are encouraging, however the tripartite model's potential to estimate distinct barrier strengths in each direction has been sacrificed and the relationships between the symmetric $T(\upsilon)$ model and the Kruuk and Gavrilets stepped cline continua remain unexplored.

The Students-t is not the only candidate distribution for expressing stepped-ness and asymmetry of clines; the generalised Logistic with parameters (1,1) matches logistic shape, where the Students-t has no exact match. The Kruuk, Gavrilets shape continua results start with that shape at their sigmoid extremes; perhaps one or the other follows the shape of the generalised logistic as they become more stepped? To explore such possibilities and decide between shape models it seems best to construct a shape space within which all the clines discussed thus far can be compared. Previous cline shape comparisons (Barton and Gale 1993; Gavrilets 1997) have focussed on zero-centred width-rescaled clines as here (*UnitClines*, though Gavrilets chose to rescale to half width), but then Logit transformed. The Logit function is the inverse of the Logistic CDF, and so the transformed Logistic (Bazykin) cline is linear, other shapes deviating from linearity (figure 2.2 in Barton and Gale 1993). However, these deviations remain difficult to interpret, and the justification for comparing all other cline shapes to Bazykin is weakened when (i) we see other clines also have simple inverses (Table 1) and (ii) we remember the Bazykin result assumes a Normal kernel. Instead, here we seek a shape comparison framework independent of all the cline shapes we wish to compare.

5.1 | A continuous model of barrier effects

The T-cline model is a re-expression of existing probability distribution results. In this section we take an approach closer to the development of the tripartite cline model: We construct a cline model using simpler functions as building blocks. This construction differs from the tripartite cline in that it is continuous, that is, without joins or 'corners'. It differs from the T-cline in that it requires two variables, not one, to parameterise stepped-ness.

Barriers to gene flow have units of distance, may be found at cline centres, and are expected to change the shape of clines (Barton and Bengtsson 1986). Suppose the effect of a barrier to gene flow is, from the gene perspective, to increase the 'subjective' distance experienced when crossing the cline centre. Box 2 shows how such a model can be developed, and Figure 3 places all the symmetric cline shapes and continua of cline shapes discussed thus far in the context of this *lamdal* barrier effect model.

The Figure 3 comparison of cline shapes serves several purposes. First, it shows that two shape parameters are sufficient to accurately model all symmetric clines we report from the literature: using the *lamdal* model developed in Box 2 for a Normal cline with T-cline distance distortion, Figure 3 fits are as good as or better than the green line fit to Fisher's data in Figure 2. Basing the Box 2 model on the notion of barriers distorting (increasing) the 'subjective' distance required for genes to cross between taxa appears justified.

Second we see all of these cline models set for the first time in a single comparative framework that includes the shapes of the Normal and Cauchy (sigmoid and stepped) neutral cline shapes of Figure 1, alongside clines under selection: Bazykin (sigmoid), Kruuk and Gavrilets (stepped continuua for single and multiple loci respectively). We can now see that the two continua follow somewhat similar trajectories in shape as their stepped-ness increases from their shared Bazykin endpoints, the difference being largely in terms of the weight λ of central spatial distortion.

Third, while the spatial population genetics continua of cline shapes cross paths with the shapes of CDFs of known probability distributions in Figure 3, we see they do not align. In the *lamdal* shape space the former run horizontally, the latter more vertically. It therefore seems unlikely that there is an oven ready CDF waiting in the literature for us to discover and use as a model that will better embody the notion of smoothly stepped clines for inference. Of the two CDF continua in Figure 3 the T-cline family appears preferable for inference, as it allows for both neutral cline shape (Normal (sigmoid) at high v) and stepped shape, whereas the generalised logistic distribution terminates at Laplace (double exponential kernel) shape. It also seems that an analytical solution unifying the Kruuk and Gavrilets continua stepped clines, and generalising this over diverse kernel shapes is not on the horizon. We can however fit the *lamdal* model to data and see where the shape parameter estimates fall in relation to these existing results (See Supporting Information).

The *lamdal* model naturally extends to the asymmetric case when we allow its central barrier *UnitCline* to be asymmetric, for example defined by an asymmetric T-cline such as in Figure 2. In fact, the model should not be used for inference *without* this possibility of asymmetry, using *lamdal* central distance distortion such as $UnitNormal\left(f_{\lambda,l}\left(\frac{x-c}{w}\right)\right)$, with $U_B = T(0.15, \alpha)$. Here $\{c, w\}$ are the cline centering and scaling parameters, $\{l, \lambda\}$ the shaping parameters, while parameter $\{\alpha\}$ allows for departures in symmetry away from the Figure 3 case. The tripartite cline requires six parameters for inference, the *lamdal* model five: $\{c, w, l, \lambda, \alpha\}$, while the T-cline model maintains flexibility for both shape and asymmetry with just four parameters: $\{c, w, v, \alpha\}$. There is a simple reason to maintain flexibility in both shape and asymmetry when simplifying cline models: to avoid mis-inference. For example, if we make an assumption of symmetry during inference, and we are wrong, then estimates of cline centres and widths can be mis-inferred (Baird and Macholan 2012, box 14.3). The issue of mis-inference due to under-parameterisation is very general, but perhaps best illustrated with respect to outlier analyses.

6 | Outlier Analyses and Genomic Clines

Suppose we infer a sigmoid symmetric cline 1 km wide as in Box 1, but the field data is insufficient to distinguish between a Normal sigmoid shape and a Bazykin (logistic) sigmoid. The data are then consistent with two very different scenarios: either

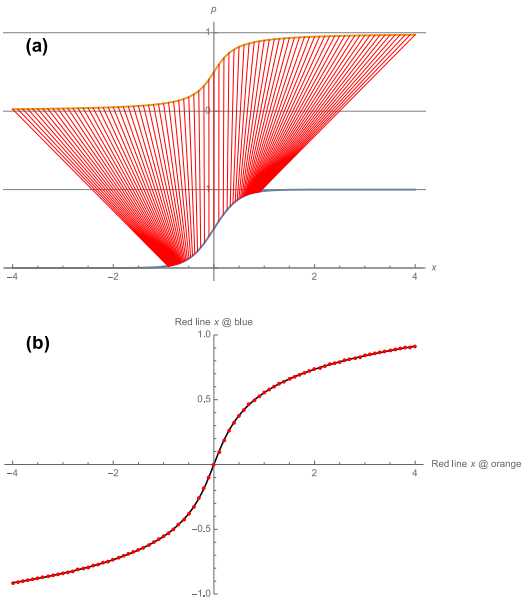
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BOX 2 | *'Lamdal'* model of continuous cline shape.

We seek a continuous approximation to a stepped unit cline, assuming a barrier distorts the gene's eye view of distance travelled during geneflow. First, exploring the exact nature of such a distortion, suppose unit cline U_N is 'sigmoid' and U_C is stepped. By the nature of inverse functions

$$U_N\left(U_N^{-1}\left(U_C(x)\right)\right) = U_C(x)$$

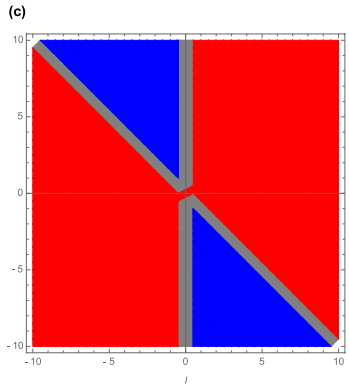
$$U_N(f(x)) = U_C(x); f(x) = U_N^{-1}\left(U_C(x)\right)$$



Here $f(x)$ is an exact distortion of distance x such that sigmoid U_N becomes stepped U_C. (a) A mapping between cline shapes. Red lines connect points of equal p on two unit clines, above *UnitCauchy*, below *UnitNormal*. (b) The exact distance distortion for the above mapping. Red points are the {Cauchy, Normal} x-coordinates of the ends of each red line in (a), the black line is $f(x)$ for these two cline shapes. Under this distance distortion *UnitNormal* becomes *UnitCauchy*. The form of $f(x)$ suggest a two parameter approximation to the distance distortion for stepped cline shapes in general:

$$f(x) \approx f_{\lambda,l}(x) = X + \lambda\left(U_B(X/l) - U_B(0)\right);$$

$$X = \frac{x}{1 - \frac{\lambda}{l}}$$



The *'lamdal'* (λ, l) distorted distance increases linearly other than with the central effects of $U_B(\cdot)$, a barrier *UnitCline* p -recentered by $U_B(0)$, rescaled by l and with effect weighted by λ. Distance is rescaled $x \to X$ such that any *UnitCline* will retain unit width under *lamdal* distortion. (c) shows the (red) space in which *lamdal* distortion of *UnitClines* results in the desired model of monotonic rising *UnitClines*. Blue regions violate these rules. The apparently simple form of the *lamdal* approximation suggests simple interpretation of $\{\lambda, l\}$, but the valid (red) space in (c), which includes negative values of both arguments (due to the $x \to X$ rescaling) belies this suggestion; the *lamdal* approximation is instead best interpreted in the context of how it fits to known cline shapes, as in Figure 3.

a 1 km cline is being maintained by Bazykin selection (of unknown strength) or a neutral cline has expanded in width to 1 km (and we do not know how long this took). If we knew the per generation dispersal scale of the organism, then we would be able to estimate the selection acting, or the time since contact, and perhaps distinguish the two scenarios. Unfortunately, the key to unlocking such evolutionary puzzles σ, the natural measure of spatial process occurring in hybrid zones, is notoriously difficult to estimate. Just as with cline studies, capture-mark-recapture studies are notoriously data hungry, and do not actually report on per generation σ, but rather within-generation movements. These two scales of movement can obviously be very different even within the same organism, for example migratory birds with high nest-site fidelity (Ruegg 2008), leaving direct estimates of σ difficult, and the scenarios we can resolve through cline fitting reduced.

Where there is poor prior knowledge of the per generation scale of dispersal, an alternative is to measure trait cline centerings and scales (widths) relative to some global estimate. Doing so for clines in allele frequencies at sites along the genome results in outlier scans analogous to Fst scans (Rieseberg et al. 1999; Payseur et al. 2004), and the approach has been used across many systems to show sex chromosomes experience stronger barriers to geneflow than autosomes. However, just as Fst confounds diversity with dispersal, making it poorly suited to detecting geneflow outliers (Charlesworth et al. 1997; Cruickshank and Hahn 2014), so clines with only centering and scaling parameters confound multiple effects, making them poorly suited to detecting genetic barrier outliers. The most obvious confounded effect is incomplete lineage sorting (ILS). With ILS we can commonly expect clines of 'typical' centering and scaling, but incomplete fixation on one side. Fitting a model that does not allow for this asymmetry will tend to misidentify cline centering and scaling as outliers despite barrier effects on the site being 'typical'. Conversely, fitting a model that does not allow for steppedness will tend to overestimate cline width, and therefore miss true barrier effect outliers. It is for these reasons that detailed comparisons of geographic clines typically fit a hierarchy of increasingly parameterized models to check whether underparameterization of cline shape is obscuring outliers (Macholán et al. 2007). The issue is perhaps seen most clearly when this approach is taken to its natural limit, and the only constraint placed on cline models is monotonicity (Macholán et al. 2008; Carneiro et al. 2013). Then, centering and central gradient (width) estimates are seen to precisely coincide across multiple

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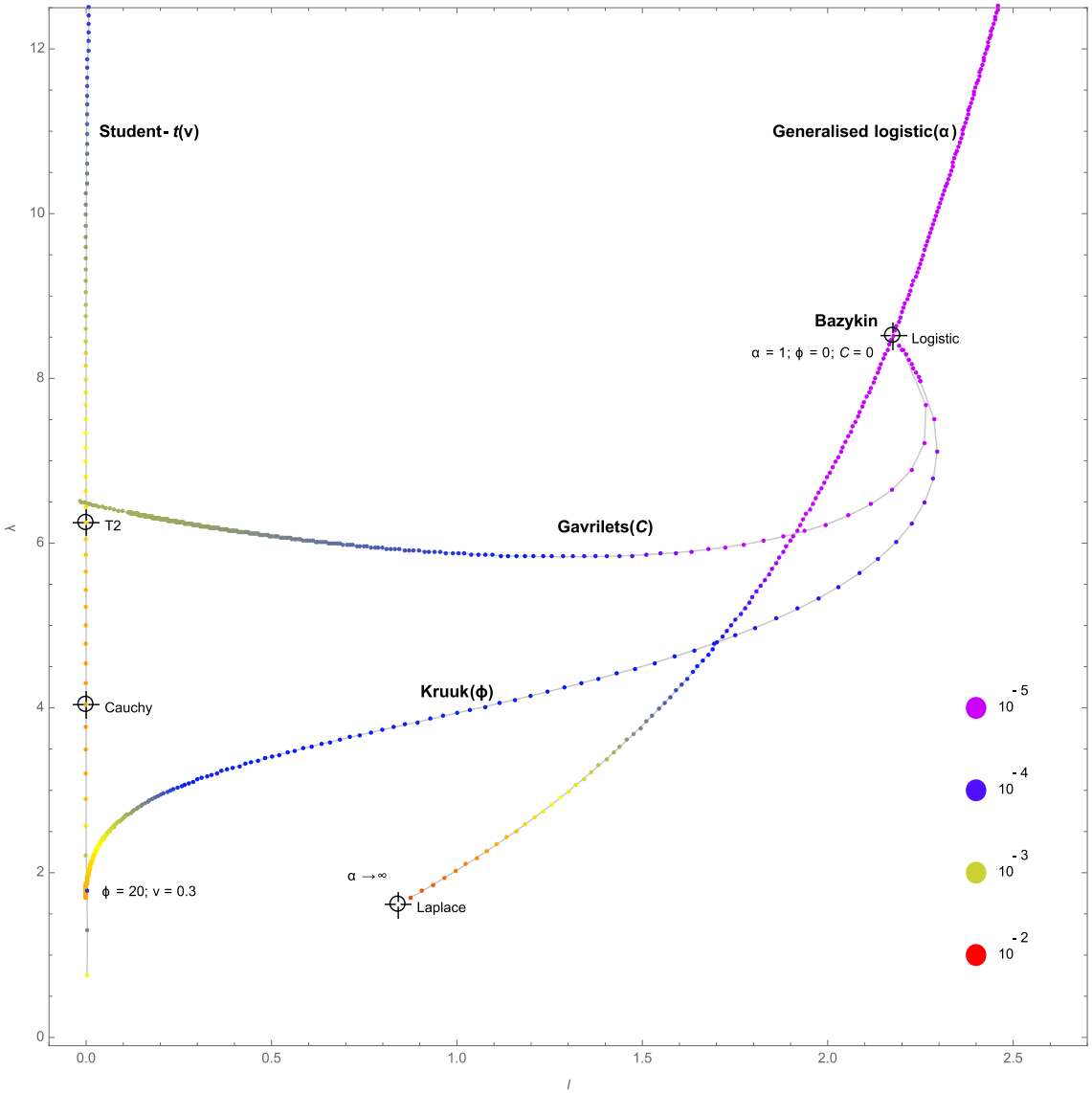


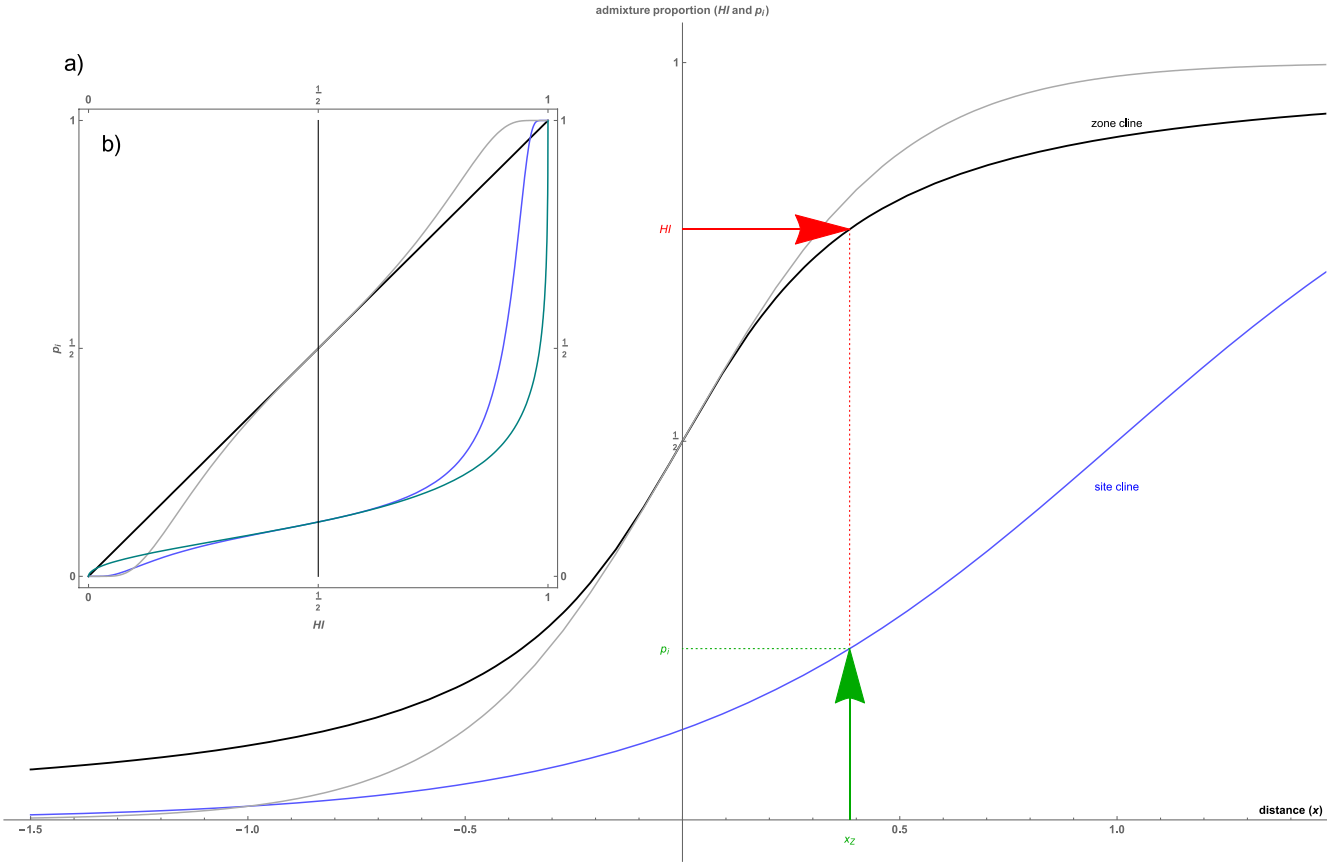
FIGURE 3 | A comparison of cline shapes. Fits to cline shapes are plotted on the plane formed by the $\{l, \lambda\}$ parameters of the *lamdal* barrier effects model $UnitNormal\left(f_{\lambda,l}(x)\right)$, with $U_B = T(0.15)$, (See Box 2). Goodness of fit is shown in colour, with a key to levels of maximum absolute difference (MAD). Sigmoid shapes are found toward the top (single locus selection against admixture, Bazykin; high v Student-t and high α generalised logistic). Stepped shapes are to the bottom and left (single locus viability selection Gavrilets with $C \to \infty$, the neutral Cauchy kernel, multilocus admixture selection Kruuk with high ϕ). The Gavrillets and Kruuk continua are shown for a series of numerical calculations for their relevant parameters. The T-cline and generalised logistic results are analytic. The *UnitNormal* shape occurs along the bottom line of the *lamdal* parameter space where the weight of central distance distortion $\lambda = 0$, and also at high v for Student-t which is by definition Normal.

genome compartments despite wildly varying asymmetric cline tails; against this background, barrier effect outliers such as the Y chromosome in figure 4a of (Macholán et al. 2008) could not be more clear.

The issues arising from under parameterization apply equally to all kinds of cline analysis, both geographic and 'genomic'. Barton suggested a 'concordance' transform (Szymura and Barton 1991), where the distance axis of a geographic cline is substituted by a global hybrid index (HI) axis. This was suggested for {centre, width} comparison of clines where geographic sampling coordinates were difficult to interpret, and has been used, for example, to compare hybrid indices of parasites and their hosts (Goüy de Bellocq et al. 2018). Barton's distance → HI axis replacement was further developed as the 'genomic clines' approach (Gompert and Buerkle 2009; Gompert and Buerkle 2012; see Macholan et al. 2011 for a comparison of genomic clines with Barton's concordance). Fitzpatrick (2013, equations 1a,b) showed a precise transformation for Bazykin's logistic geographic cline to a genomic cline and suggested this to be appropriate where "the geographic model [...] takes a more complicated form than the simple logistic function, for example, when clines are asymmetric or stepped" (Fitzpatrick 2013), presaging a surge in genomic cline fitting software (Bailey 2024; Gompert et al. 2024), but our summary of cline outlier analysis should make it clear this suggestion is over optimistic: no {centre, width}-only cline model can capture variation in cline shape and asymmetry. Box 3 illustrates how using a genomic cline model with the wrong shape can lead to mis-inference. The logit-logistic genomic cline model proposed

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BOX 3 | Genomic clines.



a. Black curve: a multilocus (genome wide) hybrid index (HI) changes across a hybrid zone. The spatial units are scaled and offset so that this is a unit cline $U_Z(x)$. Blue curve: allele frequency at site i changes as cline $U_i\left(\frac{x-c_i}{w_i}\right)$. The site cline is w_i-fold wider than the zone cline and offset by distance c_i. Grey curve: illustrating a zone cline of an alternate shape; the black curve is Cauchy, the grey logistic.
b. (inset) The curves of pane (a) when the distance axis is replaced with zone hybrid index. One further (teal) curve is shown: the blue site cline of (a) when the alternate (grey) zone cline shape is assumed.

Switching x-axes: (a) The distance x coordinate of the site cline can be replaced by the HI coordinate of the zone cline. Red arrow: applying the inverse of the zone cline to a hybrid index gives a zone x coordinate $x_Z = U_Z^{-1}(HI)$. Green arrow: applying the site cline at this x coordinate gives a site allele frequency $p_i = U_i\left(\frac{x_Z - c_i}{w_i}\right)$. Putting these two steps together, distance x is replace by HI in a generalised genomic cline

$$p_i = U_i\left(\frac{U_Z^{-1}(HI) - c_i}{w_i}\right)$$

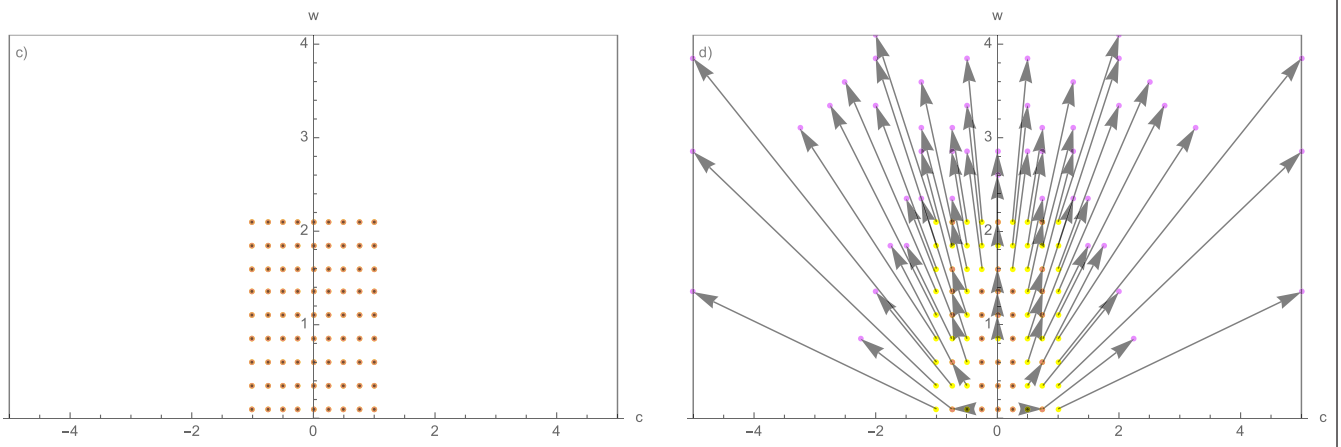
If both cline shapes $\{U_Z, U_i\}$ are logistic, the inverse U_Z^{-1} is a rescaled logit function and this is the "logit-logistic" special case genomic cline, equation (4b) of Fitzpatrick (2013), used by (Goüy de Bellocq et al. 2018), and implemented in softwares gghybrid (Bailey 2024) and bgchm (Gompert et al. 2024):

$$p_i = \frac{S^{v_i}}{S^{v_i} + (1-S)^{v_i} e^{u_i}}$$

$$S = HI; v_i = \frac{1}{w_i}; u_i = 4\frac{c_i}{w_i}; U_Z = U_i = UnitLogistic$$

Site cline shape expectations can vary along the genome, and the zone hybrid index shape may be stepped and/or asymmetric. The two parameters $\{u_i, v_i\}$ of a logit-logistic genome cline are insufficient to capture this richness of cline shapes variation.

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Misinference using the logit-logistic function: (c, d) True site cline parameter points $\{c, w\}$ are plotted in yellow, estimates of them are plotted in purple; when an estimate and a truth coincide, that point becomes brown. When estimate and truth differ, an arrow joins truth to estimate. (c) When the zone cline shape is logistic and site cline shapes are also logistic the logit-logistic function is appropriate and estimation returns the truth. (d) When the zone cline shape is instead Cauchy (as in pane (a)) but the site cline shapes remain logistic, the logit-logistic function is not appropriate for inference: estimates are only accurate when the truth is close to the origin (site cline centre, widths resemble those of the zone cline). For other site clines, errors are large and there is no simple error correction: centre and width error effects are not independent. Estimation procedure: For each true point a grid of parameter values was searched exhaustively for minimum difference between the logit-logistic function and genome cline shapes for $U_i = \mathit{UnitLogistic}$ and (c) $U_Z = \mathit{UnitLogistic}$, (d) $U_Z = \mathit{UnitCauchy}$. Eight estimations lie on the boundary of this parameter search grid, and so may represent even more extreme errors.

by Fitzpatrick (2013) is not freed of assumptions by a change in *x*-axis: in fact it implicitly assumes the shapes of *two* clines: one in (genome-wide) hybrid index, and one in allele frequency. Fitzpatrick's exact transform for the Bazykin geographic cline can, however, be generalised (Box 3) to any case where the genome-wide cline function is invertible, and Table 1 details invertible cline functions of different stepped-ness, allowing a generalised genomic cline approach to be applied assuming a variety of symmetric hybrid index cline shapes, and even stepped and asymmetric *T*-clines. These developments resolve the problem of under parameterised mis-inference over genomic cline outlier analyses in a way closely analogous to how the problem has been addressed in geographic cline analyses.

What would be the knock-on effects of extending the genomic cline model to one that allows for a better match to a globally asymmetric stepped hybrid zone such as the house mouse hybrid zone (Macholán et al. 2007)? First and foremost, any 'free lunch' impression that genomic clines allow confident estimates with little spatial sampling of a hybrid zone will likely be reduced, because fitting more parameters requires more data and may reveal confidence to be overconfidence. Second, we might expect the variance in cline centre and width estimates to be reduced, because shape mis-match effects such as those in Box 3 should be minimised. From this perspective, the shape parameters of hybrid zone clines are now acting as nuisance parameters, unused in the comparison of trait clines with each other, but nevertheless necessary if the widths and centres of those trait clines are to be comparable and estimated without overconfidence. This is not to say that the overall shape of a hybrid zone, cline or genomic wavefront is uninteresting, but rather to recognise that shape per se only becomes interesting with intensive field sampling, and that sampling has arguably become the most expensive and difficult part of such spatial analyses.

7 | Conclusions and Future Directions

Cline shapes are intimately related to the shapes of location dispersal kernels, and this allows us to draw on more than a century of work on probability distributions as a source of, and perspective on, the shapes of cline and wavefront models.

Most people find it easier to distinguish the bell-shaped curves on the left of Figure 1 than to distinguish the s-shaped curves on the right, generated from those bells. A cognitive bias against distinguishing cline shape suggests decisions regarding whether cline shapes matter should be based on objective measures, such as whether shape differences affect the results of inference (Baird and Macholan 2012, box 14.3). Here, we have seen evidence that cline shape matters not only for spatial and genomic cline inference (Figure 2, Box 2) but we are also reminded more generally that the shape of dispersal kernels matters for the speed of spatial process, and that part of the gathering phase of the coalescent during which geographically distant coalescence is improbable (Box 1).

Cline expectations are more diverse than can be usefully captured with two parameters {centre, width}; in particular, stepped and asymmetric shapes have biological importance as they can result from multilocus or non-standard forms of selection, and movement or asymmetric geneflow respectively. Further, here we have only considered the case where population density is sufficiently high that the effects of drift on cline expectations are negligible. Where drift acts it is expected to steepen the centre of site clines, while widening the overall cline in hybrid index

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(Polechová and Barton 2011), suggesting again (as with Box 3) that assuming these shapes are the same is unwise. It seems a wide variety of (stepped) symmetric shapes can be smoothly captured at high fidelity with just two shape parameters (Figure 3), and perhaps reasonably approximated with just one shape parameter—the shape parameter of the Student's-*t* distribution allows cline stepped-ness to be adjusted. The non-central Student's-*t* allows variation in both stepped-ness and asymmetry, and captures Fisher's wave of advance at high fidelity (Figure 2). It appears these continuous cline models allow better fits with fewer parameters than tripartite cline models, and computationally efficient Python tools for *T*-cline implementations are made available here: (Baird and Daley 2023). There is no free lunch: these potential advantages are at the cost of reduced interpretability of the fitting parameters. One way forward for interpretation is to fit the *lamdal* model of Figure 3 and see where estimated cline shape falls in comparison with classical models. While such developments may allow for improvements over the tripartite cline model commonly used for spatial clines, the inverse *T*-cline (Table 1) may also allow cline shape and asymmetry to be accounted for in genomic cline approaches.

Regarding our opening question: Why, after 30 years of searching, have so few stepped hybrid zones been identified? For spatial clines the answer remains unclear, but perhaps, asking simpler questions of better models with fewer parameters will allow us to decide whether stepped clines are actually rare, or just rarely proven. For genomic clines, there is a simple answer: because we were not looking; there was no genomic cline model of a stepped hybrid zone. We might hope that generalisation of genomic clines such as proposed here would allow a similar resolution of the opening question, but this is by no means obvious because the way data is sampled has changed profoundly since Haldane first fit a cline to genetic data and estimated selection (Haldane 1948). It is now extremely rare that hundreds of genetic samples are gathered from the field. Instead relatively few genomes are gathered, and from them very many nucleotide variants (SNVs) are sampled. In spatial genetics the latter does not compensate for the former, because few genomes means few spatial sampling locations, irrespective of how many SNVs are then inspected. This is equally true for genomic clines, because few genomes means few sampling points on the global HI *x*-axis. In these circumstances cline shape may be reduced to a necessary, but nuisance, parameter. A pooled sequencing approach to cline outlier analysis has recently been proposed in this context (Field et al. 2025): cline centering and width are estimated while allowing for asymmetric incomplete lineage sorting ('non-diagnostic' markers). Four degrees of freedom (in comparison to two for genomic clines) appear sufficient to clarify barrier outlier sites, and the new approach has an enormous computational advantage over classical cline fitting, making it truly suited to genomic analyses. That work, however, recognises a further issue: we cannot simply scale up cline fitting to cover (and compare) data at every one of potentially millions of SNVs—even if now computationally tractable, there are only one or two recombination events per chromosome per generation, so neighbouring sites in admixture systems are clearly not independent witnesses of the evolutionary process (Baird 2015). Assuming they are independent will lead to overconfident inference. This highlights that, to make sound admixture inference over modern genomic data, the blockwise nature of admixture tracts must be recognised (Shipilina et al. 2023; Ebdon et al. 2024). To estimate the boundaries of such tracts we should leverage every single SNV that forms a cline—their non-independence under admixture is now a positive, not a negative, and it turns out that for genomic data, introgressing blocks become obvious when SNVs are co-polarised by their association, such that all their clines are rising (or all falling) (Baird et al. 2023; Ebdon et al. 2024). The polarisation operation scales linearly with genome size and reports 'diagnostic indices', likelihood measures of the match between each SNV and a global barrier cline model estimated over individuals. This global estimate carries a *superset* of the information necessary to plot a simple cline in hybrid indices (genomes' admixtures), as it also informs us on the central bump in genomes' *heterozygosities* caused by admixture (cf p_{12} in Simon et al. 2018). Downstream inference can be targeted on regions that differ from this global estimate. If in future it becomes commonplace to genome sequence individuals sampled at very many different field co-ordinates, then cline shapes (geographic or genomic) may stop being a necessary nuisance, and start again to be of active interest. In the meantime it seems the data appetite of the questions we would wish to ask of along-genome cline outlier analyses may best be fed blockwise, and relative to change in both hybrid index and heterozygosity (cf Simon et al. 2018).

Not all modern field sampling is directed toward high throughput sequencing of genomes. Advancing technology has also increased the potential throughput of geographic locations for measures of quantitative traits and genetic markers; in particular, SNV assays such as KASP (He et al. 2014) allow for individuals from very many locations to be cost effectively assayed for scores of genetic markers (Touchard et al. 2024). Because clines in genetic and quantitative traits are governed by similar dispersal and selection processes, cline models apply equally to both (Barton and Gale 1993), and so, likewise, any developments in shaped cline models. True high throughput field sampling to take advantage of these developments risks, however, perturbing that which we wish to observe—slowing a wave of advance or narrowing a hybrid zone by reducing population density through destructive sampling. So, if there is to be a renaissance in cline inference field studies, these new models and technologies should be carefully coupled with the parallel advances that have been made in non-destructive sampling (Lefort et al. 2015).

In summary, we have presented an overview of cline theory including wavefronts, clines and inference from them. We have used dispersal kernels, as previously developed in ecology, to illustrate how clines, like statistical distributions, require more than just centering and scaling: dispersal kernel and cline shapes matter. We introduce *UnitClines* as a simple way to explore shape differences while holding centre and scale (width) constant, but they turn out to be powerful tools for constructing new cline models, both geographic and genomic. We exploit the (non central) Student-*t* distribution to provide a smooth cline model space allowing for variation in both stepped-ness and asymmetry, using half the number of shape parameters of the classic tripartite cline model. We point out such new models make more power available to infer stepped cline shape. At the same time, we note that the nature of field sampling has changed: dense sampling of a few traits has been replaced by sparse sampling of whole genomes. Cline shape still matters for outlier analyses from sparse field sampling, but shape is now used to frame nuisance parameters

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for both geographic and genomic cline models: we cannot hope for detailed shape inference, but equally we cannot pretend cline shapes and asymmetries do not vary, as then that variation becomes confounded with 'true' outlier effects on gene flow.

Author Contributions

S.J.E.B. wrote the article and the Mathematica Supporting Information. N.D. developed the Python Supporting resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Fisher's table 4: mec70109-sup-0001-DataS1.csv. **Data S2:** Fits for Figure 3: mec70109-sup-0002-DataS2.zip. **Data S3:** Mathematica CDF freely accessible through Wolfram Player.

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